

CS 6045 Advanced Algorithms

Project Proposal

Requirements: (100 points)

You need clearly mention the following components in your proposal presentation:

- 1. Title Page (10 points):** This includes the title of your project and the group members, roles and responsibilities of each member.
- 2. Statement of the Problem (10 points):** A brief background introduction of your project, describes the motivation to do the project, what is the investigated problem (formal defined). The **problem objectives** must be clearly mentioned. Usually, the objective is trying to improve the weakness of the existing algorithms.
- 3. Problem Difficulty Analysis and Proposed Method (50 points):**
 - a. Theoretic-based project: this includes NPC proof sketch of the investigated problem and a feasible solution sketch.
 - b. Programming-based project: this includes a feasible solution sketch and the explanation of why the proposed method can improve the performance compared to the latest researches.
- 4. Simulation Setting (10 points):**
 - a. Theoretic-based project: this includes the description of simulation environment, settings, and data sources, etc.
 - b. Programming-based project: this includes the details of programming environment, settings, data sources, and

metrics to evaluate and analysis your proposed algorithm, etc.

- 5. Milestones and Timeline (10 points):** include a timeline with a set of milestones for completing your project. It is recommended having a set of milestones for each week. If your group has multiple members, consider assigning roles and milestones to each of the group members.
- 6. Reference (10 points):** make sure to show authors, paper title, publication source, and publication year for each reference.

Clarity, logical presentation, completeness, elegant format, meaningful contents, readability, and typos/errors are all related to a score.

Submitting Assignment:

Submit your proposal through [Desire2Learn](#) using the appropriate link.